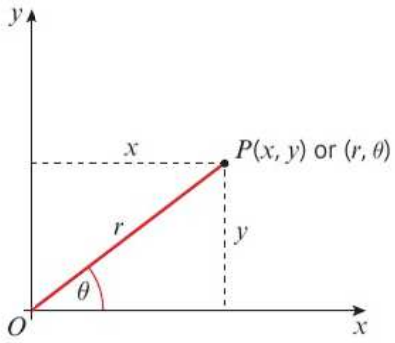
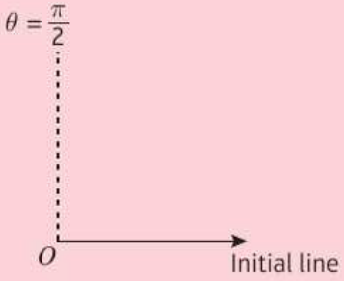
5.1 Polar coordinates and equations

Polar coordinates are an alternative way of describing the position of a point P in two-dimensional space. You need two measurements: firstly, the distance the point is from the **pole** (usually the origin O), r, and secondly, the angle measured anticlockwise from the **initial line** (usually the positive x-axis), θ. Polar coordinates are written as (r, θ).



Notation When working in polar coordinates the axes might also be labelled like this:



The coordinates of P can be written in either Cartesian form as (x, y) or in polar form as (r, θ).

You can convert between Cartesian coordinates and polar coordinates using right-angled triangle trigonometry.

From the diagram above you can see that:

- $r\cos\theta = x$
  $r\sin\theta = y$
- $r^2 = x^2 + y^2$
  $\theta = \arctan\left(\frac{y}{x}\right)$

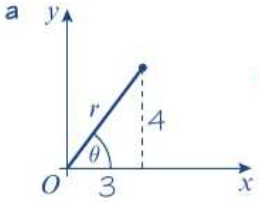
Watch out Always draw a sketch diagram to check in which quadrant the point lies, and always measure the polar angle from the positive x-axis.

Example 1

Find polar coordinates of the points with the following Cartesian coordinates.

a $(3, 4)$ **b** $(5, -12)$ **c** $(-\sqrt{3}, -1)$

a



$r = \sqrt{3^2 + 4^2} = 5$

$\theta = \arctan\frac{4}{3} = 0.927\ldots$

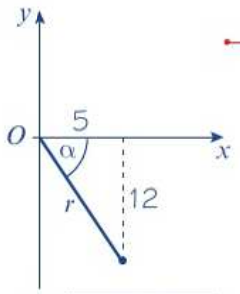
So the polar coordinates are $(5, 0.927)$

Draw a sketch.

Use Pythagoras' theorem to find r.

Use trigonometry to find θ. Give your answer in radians.

b



Draw a sketch.

$r = \sqrt{5^2 + (-12)^2} = 13$

Use Pythagoras' theorem to find r.

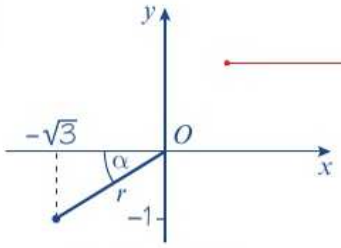
$\alpha = \arctan\frac{12}{5} = 1.176\ldots$

So $\theta = -1.176$

Use trigonometry to find θ, taking care to ensure it is in the correct quadrant. You could also write this point as (13, 5.107) since $-1.176 + 2\pi = 5.107$

So the polar coordinates are (13, −1.176)

c



The sketch shows that the point is in the 3rd quadrant.

$r = \sqrt{(-\sqrt{3})^2 + (-1)^2} = 2$

Use Pythagoras' theorem to find r.

$\alpha = \arctan\frac{1}{\sqrt{3}} = \frac{\pi}{6}$

So $\theta = \pi + \frac{\pi}{6} = \frac{7\pi}{6}$

The point is in the third quadrant so use $\theta = \frac{7\pi}{6}$
You could also use $\theta = \frac{7\pi}{6} - 2\pi = -\frac{5\pi}{6}$

So the polar coordinates are $\left(2, \frac{7\pi}{6}\right)$

Example 2

Convert the following polar coordinates into Cartesian form. The angles are measured in radians.

a $\left(10, \frac{4\pi}{3}\right)$ **b** $\left(8, \frac{2\pi}{3}\right)$

a $x = r\cos\theta = 10\cos\frac{4\pi}{3} = -5$

$y = r\sin\theta = 10\sin\frac{4\pi}{3} = -5\sqrt{3}$

So the Cartesian coordinates are $(-5, -5\sqrt{3})$

b $x = r\cos\theta = 8\cos\frac{2\pi}{3} = -4$

$y = r\sin\theta = 8\sin\frac{2\pi}{3} = 4\sqrt{3}$

So the Cartesian coordinates are $(-4, 4\sqrt{3})$

Polar equations of curves are usually given in the form $r = f(\theta)$. For example,

$r = 2\cos\theta$

$r = 1 + 2\theta$

$r = 3$ — In this example r is constant.

You can convert between polar equations of curves and their Cartesian forms.

Example 3

Find Cartesian equations of the following curves.

a $r = 5$ **b** $r = 2 + \cos 2\theta$ **c** $r^2 = \sin 2\theta, \quad 0 < \theta \leqslant \frac{\pi}{2}$

a $r = 5$

Square both sides to get $r^2 = 25$

So a Cartesian equation is $x^2 + y^2 = 25$

You need to replace r with an equation in x and y. Use $r^2 = x^2 + y^2$. So the equation $r = 5$ represents a circle with centre O and radius 5.

b $r = 2 + \cos 2\theta$

$r = 1 + (1 + \cos 2\theta)$

$r = 1 + 2\cos^2\theta$

You need an equation in x and y, so use $x = r\cos\theta$. This means first writing $\cos 2\theta$ in terms of $\cos\theta$.

Multiply by r^2:

$r^3 = r^2 + 2r^2\cos^2\theta$

Now use $x = r\cos\theta$ and $r^2 = x^2 + y^2$.

$(x^2 + y^2)^{\frac{3}{2}} = x^2 + y^2 + 2x^2$

Or $(x^2 + y^2)^{\frac{3}{2}} = 3x^2 + y^2$

Watch out Polar coordinates often give rise to complicated Cartesian equations, which cannot be written easily in the form $y = \ldots$

c $r^2 = \sin 2\theta, \quad 0 < \theta \leqslant \frac{\pi}{2}$

$r^2 = 2\sin\theta\cos\theta$

Multiply by r^2:

$r^4 = 2 \times r\sin\theta \times r\cos\theta$

$(x^2 + y^2)^2 = 2xy$

Problem-solving

You need to use the substitutions $x = r\cos\theta$ and $y = r\sin\theta$. Use $\sin 2\theta \equiv 2\sin\theta\cos\theta$ and then multiply by r^2.

Example 4

Find polar equations for the following:

a $y^2 = 4x$ **b** $x^2 - y^2 = 5$ **c** $y\sqrt{3} = x + 4$

a $y^2 = 4x$

$r^2\sin^2\theta = 4r\cos\theta$

Substitute $x = r\cos\theta$ and $y = r\sin\theta$.

$r\sin^2\theta = 4\cos\theta$

Divide by r and simplify.

$r = \dfrac{4\cos\theta}{\sin^2\theta} = 4\cot\theta\,\mathrm{cosec}\,\theta$

So a polar equation is $r = 4\cot\theta\,\mathrm{cosec}\,\theta$

b $x^2 - y^2 = 5$

$r^2\cos^2\theta - r^2\sin^2\theta = 5$

Substitute $x = r\cos\theta$ and $y = r\sin\theta$.

$r^2(\cos^2\theta - \sin^2\theta) = 5$

$r^2\cos 2\theta = 5$

Use $\cos 2\theta \equiv \cos^2\theta - \sin^2\theta$.

So a polar equation is $r^2 = 5\sec 2\theta$

c $y\sqrt{3} = x + 4$

$r\sqrt{3}\sin\theta = r\cos\theta + 4$

Substitute $x = r\cos\theta$ and $y = r\sin\theta$ and then try to simplify the trigonometric expression.

$r(\sqrt{3}\sin\theta - \cos\theta) = 4$

$r\left(\frac{\sqrt{3}}{2}\sin\theta - \frac{1}{2}\cos\theta\right) = \frac{4}{2}$

$r\sin\left(\theta - \frac{\pi}{6}\right) = 2$

Use the $\sin(A - B)$ formula.

So a polar equation is $r = 2\operatorname{cosec}\left(\theta - \frac{\pi}{6}\right)$

Exercise 5A

1 Find polar coordinates of the points with the following Cartesian coordinates.

a (5,12) **b** (−5, 12) **c** (−5, −12)

d (2, −3) **e** $(\sqrt{3}, -1)$

2 Convert the following polar coordinates into Cartesian form.

a $\left(6, \frac{\pi}{6}\right)$ **b** $\left(6, -\frac{\pi}{6}\right)$ **c** $\left(6, \frac{3\pi}{4}\right)$

d $\left(10, \frac{5\pi}{4}\right)$ **e** $(2, \pi)$

3 Find Cartesian equations for the following curves, where a is a positive constant.

a $r = 2$ **b** $r = 3\sec\theta$ **c** $r = 5\operatorname{cosec}\theta$

d $r = 4a\tan\theta\sec\theta$ **e** $r = 2a\cos\theta$ **f** $r = 3a\sin\theta$

g $r = 4(1 - \cos 2\theta)$ **h** $r = 2\cos^2\theta$ **i** $r^2 = 1 + \tan^2\theta$

4 Find polar equations for the following curves.

a $x^2 + y^2 = 16$ **b** $xy = 4$ **c** $(x^2 + y^2)^2 = 2xy$

d $x^2 + y^2 - 2x = 0$ **e** $(x + y)^2 = 4$ **f** $x - y = 3$

g $y = 2x$ **h** $y = -\sqrt{3}x + a$ **i** $y = x(x - a)$

Challenge

Show that the distance, d, between the two points (r_1, θ_1) and (r_2, θ_2) in polar coordinates is

$d = \sqrt{r_1^2 + r_2^2 - 2r_1r_2\cos(\theta_1 - \theta_2)}$

5.2 Sketching curves

You can sketch curves given in polar form by learning the shapes of some standard curves.

- **$r = a$ is a circle with centre O and radius a.**
- **$\theta = \alpha$ is a half-line through O and making an angle α with the initial line.**
- **$r = a\theta$ is a spiral starting at O.**

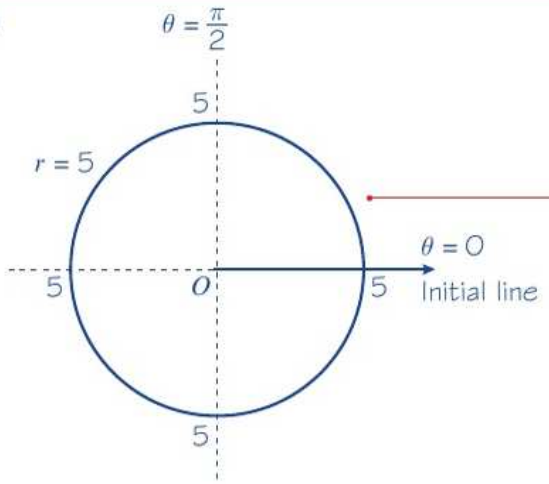
Example 5

Sketch the following curves.

a $r = 5$ **b** $\theta = \frac{3\pi}{4}$ **c** $r = a\theta$

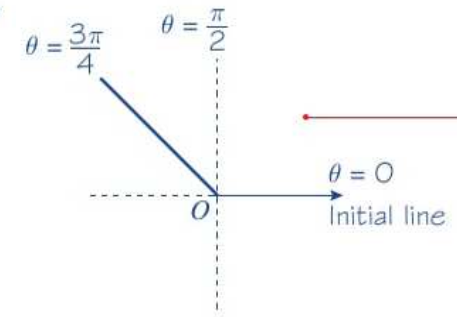
where a is a positive constant.

a



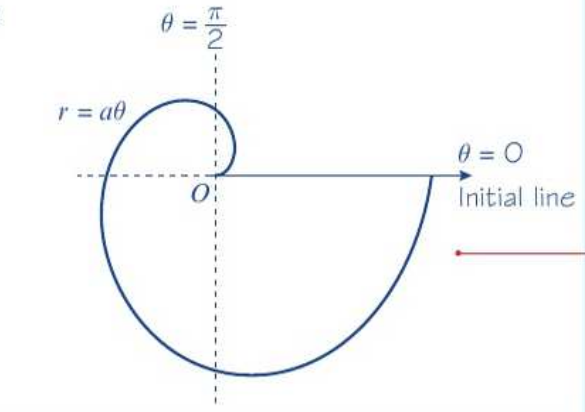
This is a standard curve: a circle with centre O and radius 5.

b



This is another standard graph: a half-line. Notice it is only 'half' of the line $y = -x$. The other half of the line would have equation $\theta = -\frac{\pi}{4}$ or $\theta = \frac{7\pi}{4}$

c



This is another standard curve: a spiral. It crosses the horizontal axis at $-a\pi$, 0 and $2a\pi$ and the vertical axis at $\frac{a\pi}{2}$ and $-\frac{3a\pi}{2}$. The curve here drawn for values of θ in the range $0 \leqslant \theta \leqslant 2\pi$.

You can also sketch curves by drawing up a table of values of r for particular values of θ.
It is common to choose only values of θ that give positive values of r.

Watch out Some graph-drawing programs and graphical calculators will sketch polar curves for negative values of r so take care when using these tools to help you.

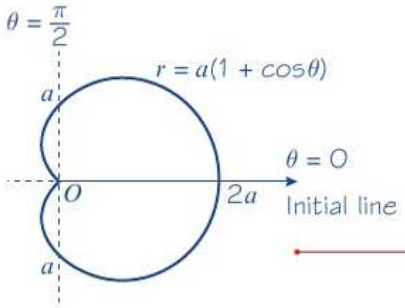
Example 6

Sketch the following curves.

a $r = a(1 + \cos\theta)$ **b** $r = a\sin 3\theta$ **c** $r^2 = a^2\cos 2\theta$

a $r = a(1 + \cos\theta)$

θ	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
r	$2a$	a	0	a	$2a$



This curve is 'heart' shaped and is known as a cardioid.

b $r = a\sin 3\theta$

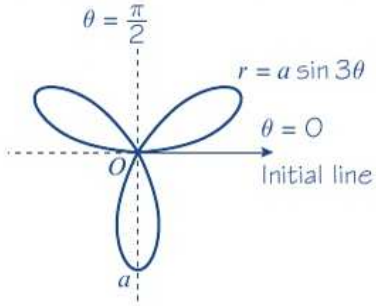
Need to consider

$0 \leqslant \theta \leqslant \frac{\pi}{3}, \frac{2\pi}{3} \leqslant \theta \leqslant \pi$ and $\frac{4\pi}{3} \leqslant \theta \leqslant \frac{5\pi}{3}$

Since we only draw the curve when $r \geqslant 0$ you need to determine the values of θ required.

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{3}$
r	0	a	0

Choose values of θ which give exact values of r. The values shown here define the first loop of the curve. The values of r will be the same in the other two loops.



c $r^2 = a^2\cos 2\theta$

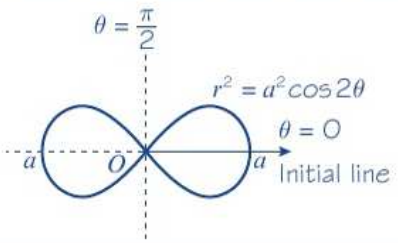
You need values of θ in the ranges

$-\frac{\pi}{4} \leqslant \theta \leqslant \frac{\pi}{4}$ and $\frac{3\pi}{4} \leqslant \theta \leqslant \frac{5\pi}{4}$

Establish the values of θ for which the curve exists.

θ	$-\frac{\pi}{4}$	0	$\frac{\pi}{4}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$
r	0	a	0	0	a	0

Draw up a table of values and sketch the curve.



Problem-solving

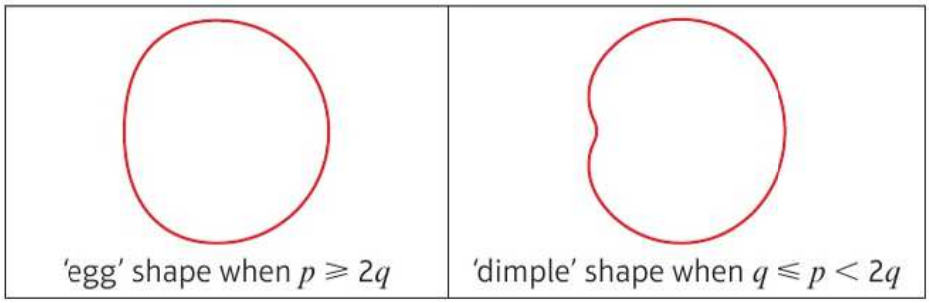
When sketching polar curves it is useful to plot points for key values of θ. Make a table of values for θ at multiples of $\frac{\pi}{2}$ to determine the points at which the curve meets or intersects the coordinate axes.

Problem-solving

The curve given by $r = a\sin 3\theta$ is typical of the patterns that arise in polar curves for equations of the form $r = a\cos n\theta$ or $r = a\sin n\theta$. They will have n loops symmetrically arranged around O.

Online Explore curves given in polar form using GeoGebra.

Curves with equations of the form $r = a(p + q\cos\theta)$ are defined for all values of θ if $p \geqslant q$. An example of this, when $p = q$, was the cardioid seen in Example 6a. These curves fall into two types, those that are 'egg' shaped (i.e. a convex curve) and those with a 'dimple' (i.e. the curve is concave at $\theta = \pi$). The conditions for each type are given below:



'egg' shape when $p \geqslant 2q$ | 'dimple' shape when $q \leqslant p < 2q$

Links You can prove these conditions by considering the number of tangents to the curve that are perpendicular to the initial line.

→ Example 14

Example 7

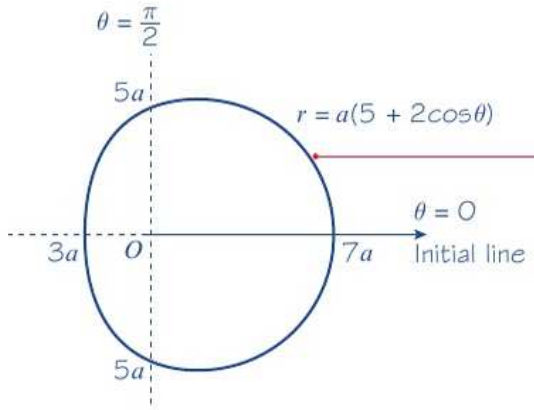
Sketch the following curves.

a $r = a(5 + 2\cos\theta)$ **b** $r = a(3 + 2\cos\theta)$

a $r = a(5 + 2\cos\theta)$

θ	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$
r	$7a$	$5a$	$3a$	$5a$

Draw up a suitable table of values.

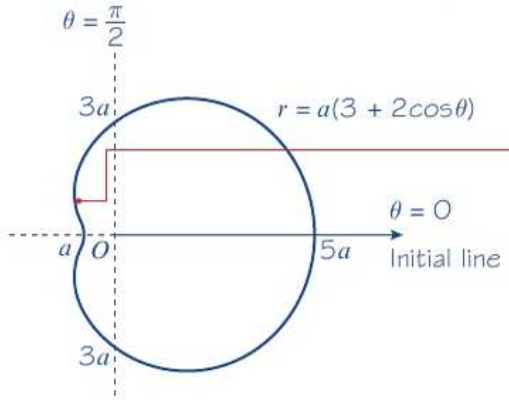


Since $5 > 2 \times 2$ there is no 'dimple'.

b $r = a(3 + 2\cos\theta)$

θ	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$
r	$5a$	$3a$	a	$3a$

Draw up a suitable table of values.



Since $3 < 2 \times 2$ there will be a 'dimple' for θ close to π.

You may also need to find a polar curve to represent a locus of points on an Argand diagram.

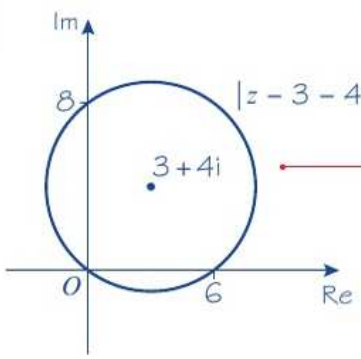
Links If the pole is taken as the origin, and the initial line is taken as the positive real axis, then the point (r, θ) will represent the complex number $re^{i\theta}$ ← Section 1.1

Example 8

a Show on an Argand diagram the locus of points given by the values of z satisfying $|z - 3 - 4i| = 5$

b Show that this locus of points can be represented by the polar curve $r = 6\cos\theta + 8\sin\theta$.

a



This locus is a circle with centre 3 + 4i and radius 5.

b In Cartesian form, $(x - 3)^2 + (y - 4)^2 = 25$

$(r\cos\theta - 3)^2 + (r\sin\theta - 4)^2 = 25$

Substitute for x and y in polar form.

$r^2\cos^2\theta - 6r\cos\theta + 9 + r^2\sin^2\theta$

$- 8r\sin\theta + 16 = 25$

$r^2(\cos^2\theta + \sin^2\theta) - 6r\cos\theta - 8r\sin\theta = 0$

$r^2 = 6r\cos\theta + 8r\sin\theta$

$r = 6\cos\theta + 8\sin\theta$

Exercise 5B

1 Sketch the following curves.

a $r = 6$ **b** $\theta = \frac{5\pi}{4}$ **c** $\theta = -\frac{\pi}{4}$

d $r = 2\sec\theta$ **e** $r = 3\operatorname{cosec}\theta$ **f** $r = 2\sec\left(\theta - \frac{\pi}{3}\right)$

g $r = a\sin\theta$ **h** $r = a(1 - \cos\theta)$ **i** $r = a\cos 3\theta$

j $r = a(2 + \cos\theta)$ **k** $r = a(6 + \cos\theta)$ **l** $r = a(4 + 3\cos\theta)$

m $r = a(2 + \sin\theta)$ **n** $r = a(6 + \sin\theta)$ **o** $r = a(4 + 3\sin\theta)$

p $r = 2\theta$ **q** $r^2 = a^2\sin\theta$ **r** $r^2 = a^2\sin 2\theta$

(E) **2** Sketch the graph with polar equation

$$r = k\sec\left(\frac{\pi}{4} - \theta\right)$$

where k is a positive constant, giving the coordinates of any points of intersection with the coordinate axes in terms of k. **(4 marks)**

E **3 a** Show on an Argand diagram the locus of points given by the values of z satisfying
$|z - 12 - 5i| = 13$ **(2 marks)**

b Show that this locus of points can be represented by the polar curve
$r = 24\cos\theta + 10\sin\theta$ **(4 marks)**

E **4 a** Show on an Argand diagram the locus of points given by the values of z satisfying
$|z + 4 + 3i| = 5$ **(2 marks)**

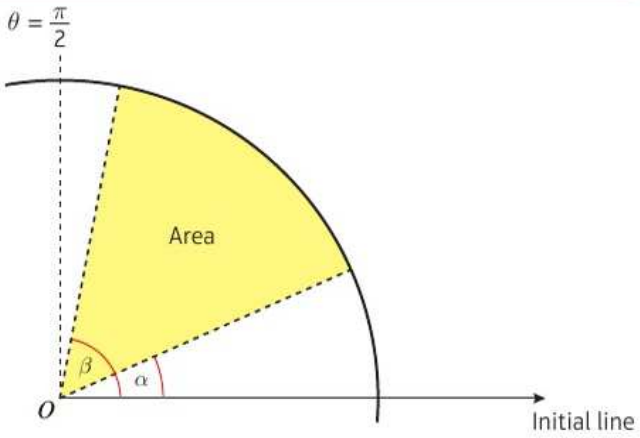
b Show that this locus of points can be represented by the polar curve
$r = -8\cos\theta - 6\sin\theta$ **(4 marks)**

5.3 Area enclosed by a polar curve

You can find areas enclosed by a polar curve using integration.

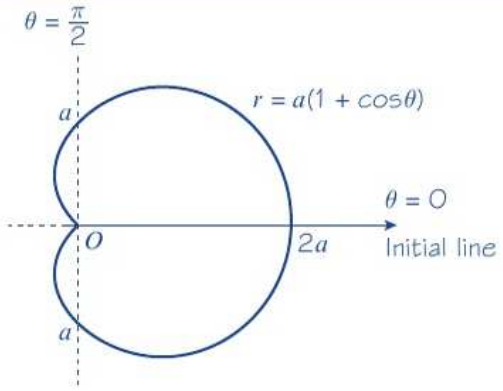
- **The area of a sector bounded by a polar curve and the half-lines $\theta = \alpha$ and $\theta = \beta$, where θ is in radians, is given by the formula**

$$\textbf{Area} = \frac{1}{2}\int_{\alpha}^{\beta} r^2 \, d\theta$$



Example 9

Find the area enclosed by the cardioid with equation $r = a(1 + \cos\theta)$.



The curve is symmetric about the initial line and so finding the area above this line and doubling it gives:

$$\text{Area} = 2 \times \frac{a^2}{2}\int_0^{\pi}(1 + \cos\theta)^2 \, d\theta$$
$$= a^2\int_0^{\pi}(1 + 2\cos\theta + \cos^2\theta)\, d\theta$$
$$= a^2\int_0^{\pi}\left(\frac{3}{2} + 2\cos\theta + \frac{1}{2}\cos 2\theta\right) d\theta$$
$$= a^2\left[\frac{3}{2}\theta + 2\sin\theta + \frac{1}{4}\sin 2\theta\right]_0^{\pi}$$
$$= a^2\left(\left(\frac{3}{2}\pi + 0 + 0\right) - 0\right)$$
$$= \frac{3a^2\pi}{2}$$

Problem-solving

Start by sketching the curve. You can simplify your calculation by using the fact that the curve is symmetric about the initial line. Hence you can integrate from 0 to π and then double your answer.

Use the formula for area. Remember to square the expression for r.

You can use trigonometric identities for $\cos 2\theta$ to integrate terms in $\cos^2\theta$ or $\sin^2\theta$:

$\cos^2\theta \equiv \frac{1 + \cos 2\theta}{2}$ ← **Pure Year 2, Chapter 11**

Watch out Unlike Cartesian integration, areas in the third and fourth quadrants do not produce negative integrals. You could obtain the same result by integrating between 0 and 2π:

$$\frac{1}{2}\int_0^{2\pi} a^2(1 + \cos\theta)^2 \, d\theta = \frac{3a^2\pi}{2}$$

Example 10

Find the area of one loop of the curve with polar equation $r = a\sin 4\theta$.

$r = a\sin 4\theta$ will have one loop for

$0 \leqslant \theta \leqslant \frac{\pi}{4}$

Find the values of θ which will give the beginning and end of a loop by solving $r = 0$.

Area $= \frac{1}{2}\int_0^{\frac{\pi}{4}} r^2\,d\theta = \frac{a^2}{2}\int_0^{\frac{\pi}{4}} \sin^2 4\theta\,d\theta$

Use the area formula.

$= \frac{a^2}{4}\int_0^{\frac{\pi}{4}} (1 - \cos 8\theta)\,d\theta$

Use the trigonometric identity for $\cos 2\theta$. In this case, $\sin^2 4\theta \equiv \frac{1 - \cos 8\theta}{2}$

$= \frac{a^2}{4}\left[\theta - \frac{\sin 8\theta}{8}\right]_0^{\frac{\pi}{4}}$

$= \frac{a^2}{4}\left(\frac{\pi}{4} - \frac{\sin 2\pi}{8}\right) - 0$

Remember $\sin 2\pi = 0$.

$= \frac{a^2\pi}{16}$

Online Explore the area enclosed by a loop of the polar curve with the form $r = a\sin\theta$ using GeoGebra.



Watch out $r = \sin n\theta$ has n loops and so a simple way of finding the area of one loop would appear to be to find $\frac{1}{2}\int_0^{2\pi} r^2\,d\theta$ and divide by n. This would give $\frac{a^2\pi}{8}$

The reason why this is not the correct answer is because when you take r^2 in the integral you are also including the n loops given by $r < 0$. You need to choose your limits carefully so that $r \geqslant 0$ for all values within the range of the integral.

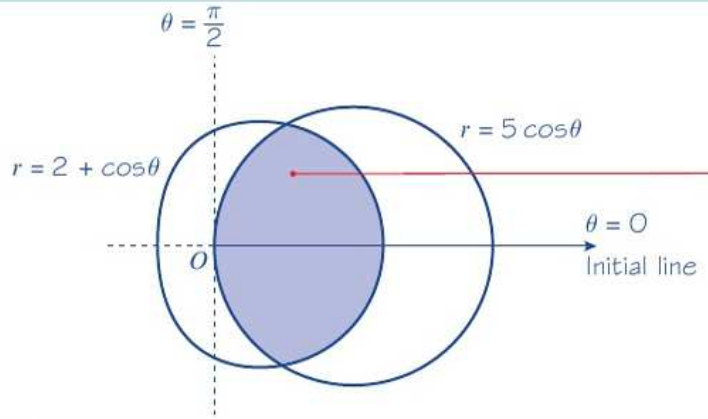
Example 11

a On the same diagram, sketch the curves with equations $r = 2 + \cos\theta$ and $r = 5\cos\theta$.

b Find the polar coordinates of the points of intersection of these two curves.

c Find the exact area of the region which lies within both curves.

a



A table of values would consider $\theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$

This is the region required in part **c**.

b The points of intersection are given by $2 + \cos\theta = 5\cos\theta$

Form a suitable equation to find the points of intersection.

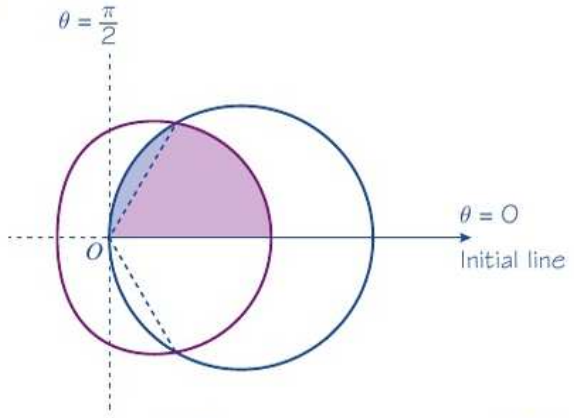
So $4\cos\theta = 2$

$\theta = \pm\frac{\pi}{3}$

So the polar coordinates are $\left(\frac{5}{2}, \pm\frac{\pi}{3}\right)$

Solve for θ and then substitute in $r = 5\cos\theta$ to find the value of r.

c



$$\text{Area} = 2 \times \tfrac{1}{2}\int_0^{\frac{\pi}{3}}(2+\cos\theta)^2\,d\theta + 2 \times \tfrac{1}{2}\int_{\frac{\pi}{3}}^{\frac{\pi}{2}}(5\cos\theta)^2\,d\theta$$

$$= \int_0^{\frac{\pi}{3}}(4 + 4\cos\theta + \cos^2\theta)\,d\theta + \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} 25\cos^2\theta\,d\theta$$

$$= \int_0^{\frac{\pi}{3}}\left(\frac{9}{2} + 4\cos\theta + \frac{\cos 2\theta}{2}\right)d\theta + \int_{\frac{\pi}{3}}^{\frac{\pi}{2}}\frac{25}{2}(1+\cos 2\theta)\,d\theta$$

$$= \left[\frac{9}{2}\theta + 4\sin\theta + \frac{\sin 2\theta}{4}\right]_0^{\frac{\pi}{3}} + \frac{25}{2}\left[\theta + \frac{\sin 2\theta}{2}\right]_{\frac{\pi}{3}}^{\frac{\pi}{2}}$$

$$= \left(\frac{3\pi}{2} + 2\sqrt{3} + \frac{\sqrt{3}}{8}\right) - (0) + \left(\frac{25\pi}{4} + 0\right) - \left(\frac{25\pi}{6} + \frac{25\sqrt{3}}{8}\right)$$

$$= \frac{43\pi}{12} - \sqrt{3}$$

Remember that the area formula gives areas of *sectors*. So you need the sector formed by the purple curve and the sector formed by the blue curve. Again you can use symmetry about the initial line.

Square and use the trigonometric identity for $\cos 2\theta$.

Use the exact value of $\sin\frac{2\pi}{3}$

Exercise 5C

1 Find the area of the finite region bounded by the curve with the given polar equation and the half-lines $\theta = \alpha$ and $\theta = \beta$.

a $r = a\cos\theta,\ \alpha = 0,\ \beta = \frac{\pi}{2}$ **b** $r = a(1+\sin\theta),\ \alpha = -\frac{\pi}{2},\ \beta = \frac{\pi}{2}$ **c** $r = a\sin 3\theta,\ \alpha = \frac{\pi}{6},\ \beta = \frac{\pi}{4}$

d $r^2 = a^2\cos 2\theta,\ \alpha = 0,\ \beta = \frac{\pi}{4}$ **e** $r^2 = a^2\tan\theta,\ \alpha = 0,\ \beta = \frac{\pi}{4}$ **f** $r = 2a\theta,\ \alpha = 0,\ \beta = \pi$

g $r = a(3+2\cos\theta),\ \alpha = 0,\ \beta = \frac{\pi}{2}$

2 Show that the area enclosed by the curve with polar equation $r = a(p + q\cos\theta)$ is $\frac{2p^2+q^2}{2}\pi a^2$.

3 Find the area of a single loop of the curve with equation $r = a\cos 3\theta$.

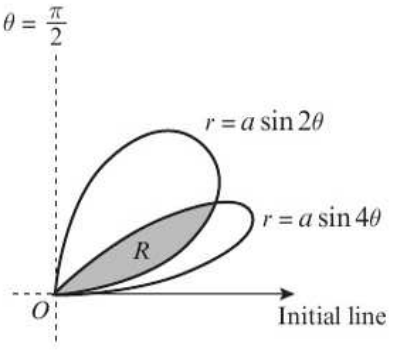
(E) **4** A curve has equation $r = a + 5\sin\theta$, $a > 5$. The area enclosed by the curve is $\frac{187\pi}{2}$

Find the value of a. **(5 marks)**

(E/P) **5** The diagram shows the curves with equations $r = a\sin 4\theta$ and $r = a\sin 2\theta$ for $0 \leqslant \theta \leqslant \frac{\pi}{2}$

The finite region R is contained within both curves.

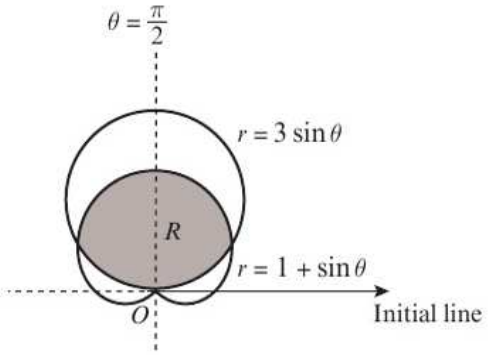
Find the area of R, giving your answer in terms of a. **(8 marks)**



E/P **6** The diagram shows the curves with equations $r = 1 + \sin\theta$ and $r = 3\sin\theta$.

The finite region R is contained within both curves.

Find the area of R. **(8 marks)**



E/P **7** The set of points, A, is defined by

$$A = \left\{z : -\frac{\pi}{4} \leqslant \arg z \leqslant 0\right\} \cap \{z : |z - 4 + 3\mathrm{i}| \leqslant 5\}$$

a Sketch on an Argand diagram the set of points, A. **(4 marks)**

Given that the locus of points given by the values of z satisfying $|z - 4 + 3\mathrm{i}| = 5$ can be expressed in polar form using the equation $r = 8\cos\theta - 6\sin\theta$,

b find, correct to three significant figures, the area of the region defined by A. **(8 marks)**

E/P **8** The set of points, A, is defined by

$$A = \left\{z : \frac{\pi}{2} \leqslant \arg z \leqslant \pi\right\} \cap \{z : |z + 12 - 5\mathrm{i}| \leqslant 13\}$$

a Sketch on an Argand diagram the set of points, A. **(4 marks)**

b Find, correct to three significant figures, the area of the region defined by A. **(8 marks)**

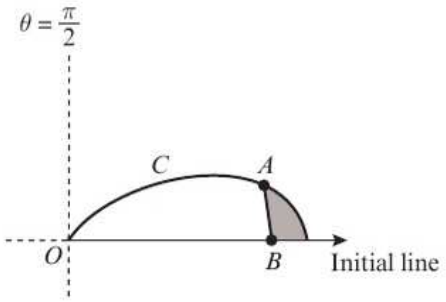
E/P **9** The diagram shows the curve C with polar equation

$r = 1 + \cos 3\theta$, $0 \leqslant \theta \leqslant \frac{\pi}{3}$

At points A and B, the value of r is $\frac{2 + \sqrt{2}}{2}$

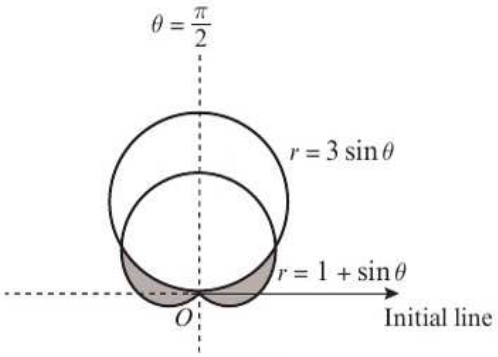
Point A lies on C and point B lies on the initial line.

Find, correct to three significant figures, the finite area bounded by the curve, the line segment AB and the initial line, shown shaded in the diagram. **(9 marks)**



E/P **10** The diagram shows the curves $r = 1 + \sin\theta$ and $r = 3\sin\theta$.

Find the shaded area, giving your answer correct to two decimal places. **(8 marks)**

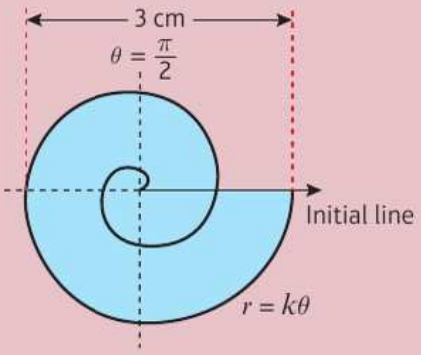


Challenge

The cross-section of a shell is modelled using the curve with polar equation $r = k\theta$, $0 \leqslant \theta \leqslant 4\pi$, where k is a positive constant. The horizontal diameter of the shell, as shown in the diagram, is 3 cm.

a Find the exact value of k.

b Hence find the total shaded area of the cross-section.



5.4 Tangents to polar curves

If you are given a curve $r = f(\theta)$ in polar form, you can write it as a parametric curve in Cartesian form, using θ as the parameter:

$$x = r\cos\theta = f(\theta)\cos\theta$$
$$y = r\sin\theta = f(\theta)\sin\theta$$

By differentiating parametrically, you can find the gradient of the curve at any point:

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$$

When $\frac{dy}{d\theta} = 0$, a tangent to the curve will be horizontal.

When $\frac{dx}{d\theta} = 0$, a tangent to the curve will be vertical.

You need to be able to find tangents to a polar curve that are **parallel** or **perpendicular** to the initial line.

- **To find a tangent parallel to the initial line set $\frac{dy}{d\theta} = 0$.**
- **To find a tangent perpendicular to the initial line set $\frac{dx}{d\theta} = 0$.**

Example 12

Find the coordinates of the points on $r = a(1 + \cos\theta)$ where the tangents are parallel to the initial line $\theta = 0$.

$y = r\sin\theta = a(\sin\theta + \sin\theta\cos\theta)$

$\frac{dy}{d\theta} = a(\cos\theta + \cos^2\theta - \sin^2\theta)$

Find an expression for y and then solve $\frac{dy}{d\theta} = 0$.

So $0 = 2\cos^2\theta + \cos\theta - 1$

$0 = (2\cos\theta - 1)(\cos\theta + 1)$

$\cos\theta = \frac{1}{2} \Rightarrow \theta = \pm\frac{\pi}{3}$

so $r = a(1 + \frac{1}{2}) = \frac{3a}{2}$

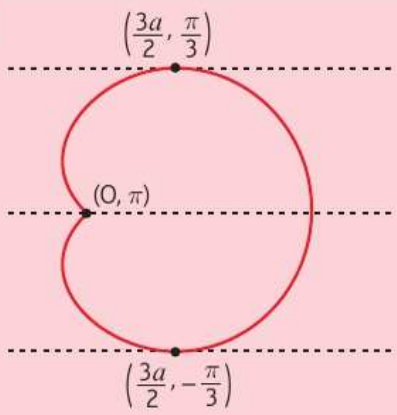
$\cos\theta = -1 \Rightarrow \theta = \pi$, and so $r = 0$

Solve the equations to find θ and then substitute back to find r.

So the tangents parallel to the initial line are at $\left(\frac{3a}{2}, \pm\frac{\pi}{3}\right)$ and $(0, \pi)$.

Problem-solving

You can see these tangents on a sketch of $y = a(1 + \cos\theta)$



Example 13

Find the equations and the points of contact of the tangents to the curve $r = a\sin 2\theta$, $0 \leqslant \theta \leqslant \frac{\pi}{2}$ that are:

a parallel to the initial line **b** perpendicular to the initial line.

Give answers to three significant figures where appropriate.

a $y = r\sin\theta = a\sin\theta\sin 2\theta$

Form an expression for y and differentiate using the product rule.

$\frac{dy}{d\theta} = a(\cos\theta\sin 2\theta + 2\cos 2\theta\sin\theta)$

$= 2a\sin\theta(\cos^2\theta + \cos^2\theta - \sin^2\theta)$

Use $\sin 2\theta \equiv 2\sin\theta\cos\theta$ and then take out the common factor. Then use $\cos 2\theta \equiv \cos^2\theta - \sin^2\theta$

$\frac{dy}{d\theta} = 0 \Rightarrow \sin\theta = 0 \Rightarrow \theta = 0$

or $2\cos^2\theta = \sin^2\theta \Rightarrow \tan\theta = \pm\sqrt{2}$

$\Rightarrow \theta = 0.955$

Choose values of θ within the range given in the question.

So $\theta = 0$ or 0.955

and $r = 0$ or $r = 2a \times \frac{\sqrt{2}}{\sqrt{3}} \times \frac{1}{\sqrt{3}}$

If $\tan a = \sqrt{2}$ then drawing a triangle shows that $\sin\alpha = \frac{\sqrt{2}}{\sqrt{3}}$ and $\cos\alpha = \frac{1}{\sqrt{3}}$

Use $\sin 2A \equiv 2\sin A\cos A$ to find r.

So the points are $(0, 0)$ and $\left(\frac{2a\sqrt{2}}{3}, 0.955\right)$

The equation of the initial line is $\theta = 0$ and that is the tangent through $(0, 0)$.

The equation of the tangent through $\left(\frac{2a\sqrt{2}}{3}, 0.955\right)$

is $y = \frac{2a\sqrt{2}}{3} \times \sin\theta = \frac{2a\sqrt{2}}{3} \times \frac{\sqrt{2}}{\sqrt{3}} = \frac{4a}{3\sqrt{3}}$

So the equation of the tangent is

$r = \frac{4a}{3\sqrt{3}}\operatorname{cosec}\theta$

Use $y = r\sin\theta$ to find the equation of the tangent and write it in polar form using $r = y\operatorname{cosec}\theta$.

b $x = r\cos\theta = a\cos\theta\sin 2\theta$

Form an expression for x and differentiate using the product rule.

$\frac{dx}{d\theta} = -a\sin\theta\sin 2\theta + 2a\cos\theta\cos 2\theta$

$= 2a\cos\theta\,(-\sin^2\theta + \cos^2\theta - \sin^2\theta)$

Use $\sin 2\theta = 2\sin\theta\cos\theta$ and then take out the common factor. Then use a formula for $\cos 2\theta$.

$\frac{dx}{d\theta} = 0 \Rightarrow \cos\theta = 0 \Rightarrow \theta = \frac{\pi}{2}$

So the y-axis is a tangent.

Or $\cos^2\theta - 2\sin^2\theta = 0 \Rightarrow \tan\theta = \pm\frac{1}{\sqrt{2}}$

If $\alpha = \frac{1}{\sqrt{2}}$ then drawing a triangle shows that $\cos\alpha = \frac{\sqrt{2}}{\sqrt{3}}$ and $\sin\alpha = \frac{1}{\sqrt{3}}$.

Use $\sin 2A \equiv 2\sin A\cos A$ to find r.

So $\theta = 0.615$

and $r = 2a \times \frac{\sqrt{2}}{\sqrt{3}} \times \frac{1}{\sqrt{3}} = \frac{2a\sqrt{2}}{3}$

The tangent is at $\left(\frac{2a\sqrt{2}}{3}, 0.615\right)$

$x = \frac{2a\sqrt{2}}{3} \times \cos\alpha = \frac{2a\sqrt{2}}{3} \times \frac{\sqrt{2}}{\sqrt{3}} = \frac{4a}{3\sqrt{3}}$

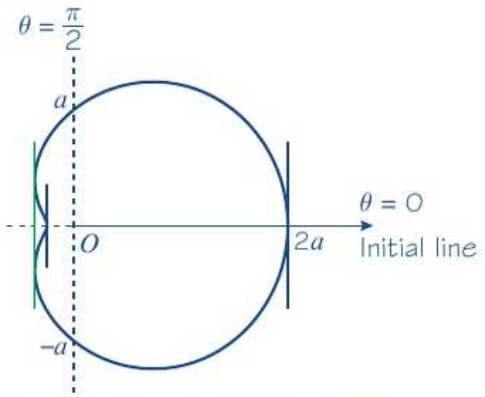
So the equation of the tangent is:

$r = \frac{4a}{3\sqrt{3}}\sec\theta$

Use $x = r\cos\theta$ to find the equation of the tangent in the form $r = x\sec\theta$.

Example 14

The curve C has equation $r = (p + q\cos\theta)$, where p and q are positive constants and $p > q$. Prove that the curve is convex for $p \geqslant 2q$, and has a dimple for $p < 2q$.



$x = r\cos\theta = p\cos\theta + q\cos^2\theta$

$\frac{dx}{d\theta} = 0 \Rightarrow 0 = -p\sin\theta - 2q\cos\theta\sin\theta$

$\Rightarrow 0 = -\sin\theta(p + 2q\cos\theta)$

This has solutions

$\sin\theta = 0$ when $\theta = 0$ or π

and $\cos\theta = -\frac{p}{2q}$

If $p < 2q$ then there will be two solutions to this equation in the second and third quadrants (the green tangents). In this case the curve is not convex and has a dimple.

If $p = 2q$ then the solution is $\theta = \pi$ and so there are only two tangents (the blue ones). In this case the curve is convex.

If $p > 2q$ then there is no solution to this equation and only the two blue tangents are possible. In this case the curve is convex.

Hence the curve is convex for $p \geqslant 2q$, and has a dimple for $p < 2q$.

Problem-solving

If the curve is not convex then there will be more than two tangents to the curve that are perpendicular to the initial line.

Find an expression for x and differentiate.

Solve the equation and consider all possible cases.

The two tangents at the two points represented by these solutions have the same equation.

Exercise 5D

1 Find the points on the cardioid $r = a(1 + \cos\theta)$ where the tangents are perpendicular to the initial line.

2 Find the points on the spiral $r = e^{2\theta}$, $0 \leqslant \theta \leqslant \pi$, where the tangents are

a perpendicular to the initial line **b** parallel to the initial line.

Give your answers to three significant figures.

3 a Find the points on the curve $r = a\cos 2\theta$, $-\frac{\pi}{4} \leqslant \theta \leqslant \frac{\pi}{4}$, where the tangents are parallel to the initial line, giving your answers to three significant figures where appropriate.

b Find the equations of these tangents.

(E) **4** Find the points on the curve with equation $r = a(7 + 2\cos\theta)$ where the tangents are parallel to the initial line. **(6 marks)**

(E) **5** Find the equations of the tangents to $r = 2 + \cos\theta$ that are perpendicular to the initial line. **(6 marks)**

(E) **6** Find the point on the curve with equation $r = a(1 + \tan\theta)$, $0 \leqslant \theta < \frac{\pi}{2}$, where the tangent is perpendicular to the initial line. **(6 marks)**

(E/P) **7** The curve C has polar equation

$$r = 1 + 3\cos\theta, \qquad 0 \leqslant \theta \leqslant \frac{\pi}{2}$$

The tangent to C at a point A on the curve is parallel to the initial line.

Point O is the pole.

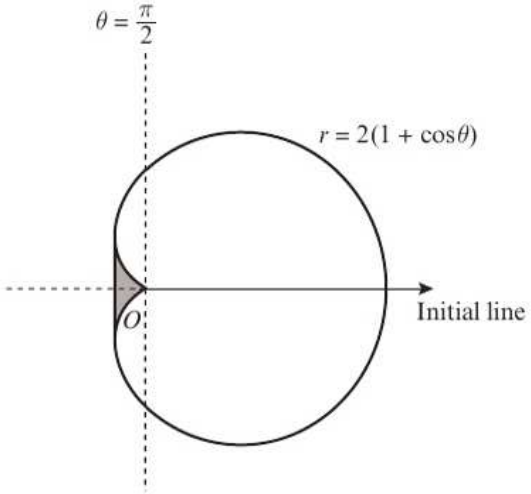
Find the exact length of the line OA. **(7 marks)**

(E/P) **8** The diagram shows a cardioid with polar equation

$r = 2(1 + \cos\theta)$

The shaded area is enclosed by the curve and the vertical line segment which is tangent to the curve and perpendicular to the initial line.

Find the shaded area, correct to three significant figures. **(8 marks)**



Mixed exercise 5

(E) **1** Determine the area enclosed by the curve with equation

$$r = a(1 + \tfrac{1}{2}\sin\theta), \quad a > 0, \quad 0 \leqslant \theta < 2\pi,$$

giving your answer in terms of a and π. **(6 marks)**

(E/P) **2 a** Sketch the curve with equation $r = a(1 + \cos\theta)$ for $0 \leqslant \theta \leqslant \pi$, where $a > 0$. **(2 marks)**

b Sketch also the line with equation $r = 2a\sec\theta$ for $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$, on the same diagram. **(2 marks)**

c The half-line with equation $\theta = \alpha$, $0 < \alpha < \frac{\pi}{2}$, meets the curve at A and the line with equation $r = 2a\sec\theta$ at B. If O is the pole, find the value of $\cos\alpha$ for which $OB = 2OA$. **(5 marks)**

(E/P) **3** Sketch, in the same diagram, the curves with equations $r = 3\cos\theta$ and $r = 1 + \cos\theta$ and find the area of the region lying inside both curves. **(9 marks)**

E **4** Find the polar coordinates of the points on $r^2 = a^2 \sin 2\theta$ where the tangent is perpendicular to the initial line. **(7 marks)**

E/P **5** **a** Shade the region R for which the polar coordinates r, θ satisfy

$$r \leqslant 4\cos 2\theta \quad \text{for} \quad -\frac{\pi}{4} \leqslant \theta \leqslant \frac{\pi}{4}$$ **(2 marks)**

b Find the area of R. **(5 marks)**

E **6** Sketch the curve with polar equation $r = a(1 - \cos\theta)$, where $a > 0$, stating the polar coordinates of the point on the curve at which r has its maximum value. **(5 marks)**

E/P **7** **a** On the same diagram, sketch the curve C_1 with polar equation

$$r = 2\cos 2\theta, \quad -\frac{\pi}{4} < \theta \leqslant \frac{\pi}{4}$$

and the curve C_2 with polar equation $\theta = \frac{\pi}{12}$ **(3 marks)**

b Find the area of the smaller region bounded by C_1 and C_2. **(6 marks)**

E **8** **a** Sketch on the same diagram the circle with polar equation $r = 4\cos\theta$ and the line with polar equation $r = 2\sec\theta$. **(4 marks)**

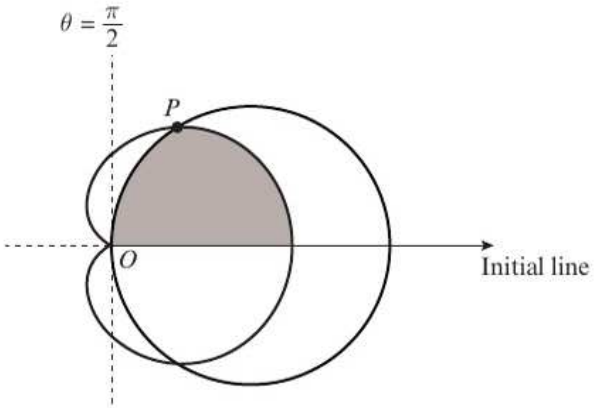
b State polar coordinates for their points of intersection. **(4 marks)**

E/P **9** The diagram shows a sketch of the curves with polar equations

$$r = a(1 + \cos\theta) \text{ and } r = 3a\cos\theta, \ a > 0$$

a Find the polar coordinates of the point of intersection P of the two curves. **(4 marks)**

b Find the area, shaded in the figure, bounded by the two curves and by the initial line $\theta = 0$, giving your answer in terms of a and π. **(7 marks)**



E/P **10** Obtain a Cartesian equation for the curve with polar equation

a $r^2 = \sec 2\theta$ **(4 marks)**

b $r^2 = \operatorname{cosec} 2\theta$ **(4 marks)**

E/P **11** **a** Show on an Argand diagram the locus of points given by the values of z satisfying

$$|z - 1 - \mathrm{i}| = \sqrt{2}$$ **(2 marks)**

b Show that this locus of points can be represented by the polar curve

$$r = 2\cos\theta + 2\sin\theta$$ **(4 marks)**

The set of points, A, is defined by

$$A = \left\{ z : \frac{\pi}{6} \leqslant \arg z \leqslant \frac{\pi}{2} \right\} \cap \{ z : |z - 1 - \mathrm{i}| \leqslant \sqrt{2} \}$$

c Show, by sketching on your Argand diagram, the set of points, A. **(2 marks)**

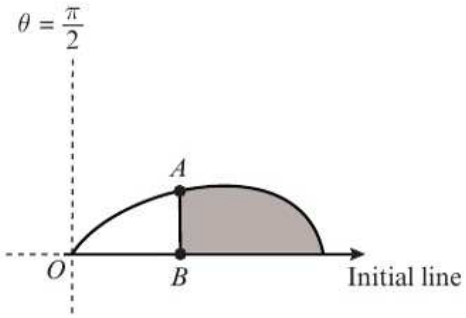
d Find, correct to three significant figures, the area of the region defined by A. **(5 marks)**

(E) **12** The diagram shows the curve C with polar equation

$$r = 4\cos 2\theta, \qquad 0 \leqslant \theta \leqslant \frac{\pi}{4}$$

At point A the value of r is 2. Point A lies on C and point B lies on the initial line vertically below A.

Find, correct to three significant figures, the area of the finite region bounded by the curve, the line segment AB and the initial line, shown shaded in the diagram. **(9 marks)**

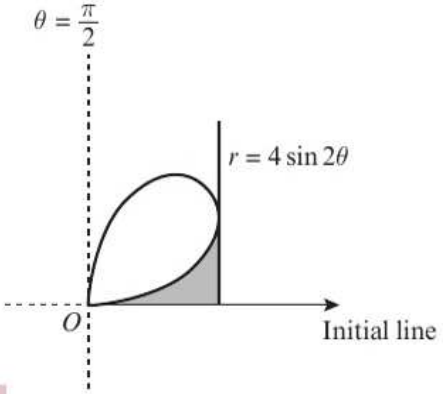


(E/P) **13** The diagram shows the curve with polar equation

$$r = 4\sin 2\theta, \qquad 0 \leqslant \theta \leqslant \frac{\pi}{2}$$

The shaded region is bounded by the curve, the initial line and the tangent to the curve which is perpendicular to the initial line.

Find, correct to two decimal places, the area of the shaded region. **(8 marks)**



Challenge

The curve C has polar equation $r = \sqrt{2}\theta$.
Show that an equation for the tangent to the curve at the point where $\theta = \frac{\pi}{4}$ is $2(\pi - 4)y + 2(\pi + 4)x = \pi^2$

Summary of key points

1 For a point P with polar coordinates (r, θ) and Cartesian coordinates (x, y),

- $r\cos\theta = x$ and $r\sin\theta = y$
- $r^2 = x^2 + y^2$, $\theta = \arctan\left(\frac{y}{x}\right)$

Care must be taken to ensure that θ is in the correct quadrant.

2
- $r = a$ is a circle with centre O and radius a.
- $\theta = \alpha$ is a half-line through O and making an angle α with the initial line.
- $r = a\theta$ is a spiral starting at O.

3 The **area of a sector** bounded by a polar curve and the half-lines $\theta = \alpha$ and $\theta = \beta$, where θ is in radians, is given by the formula

$$\textbf{Area} = \tfrac{1}{2}\int_{\alpha}^{\beta} r^2 \, \mathrm{d}\theta$$

4
- To find a tangent parallel to the initial line set $\frac{\mathrm{d}y}{\mathrm{d}\theta} = 0$.
- To find a tangent perpendicular to the initial line set $\frac{\mathrm{d}x}{\mathrm{d}\theta} = 0$.

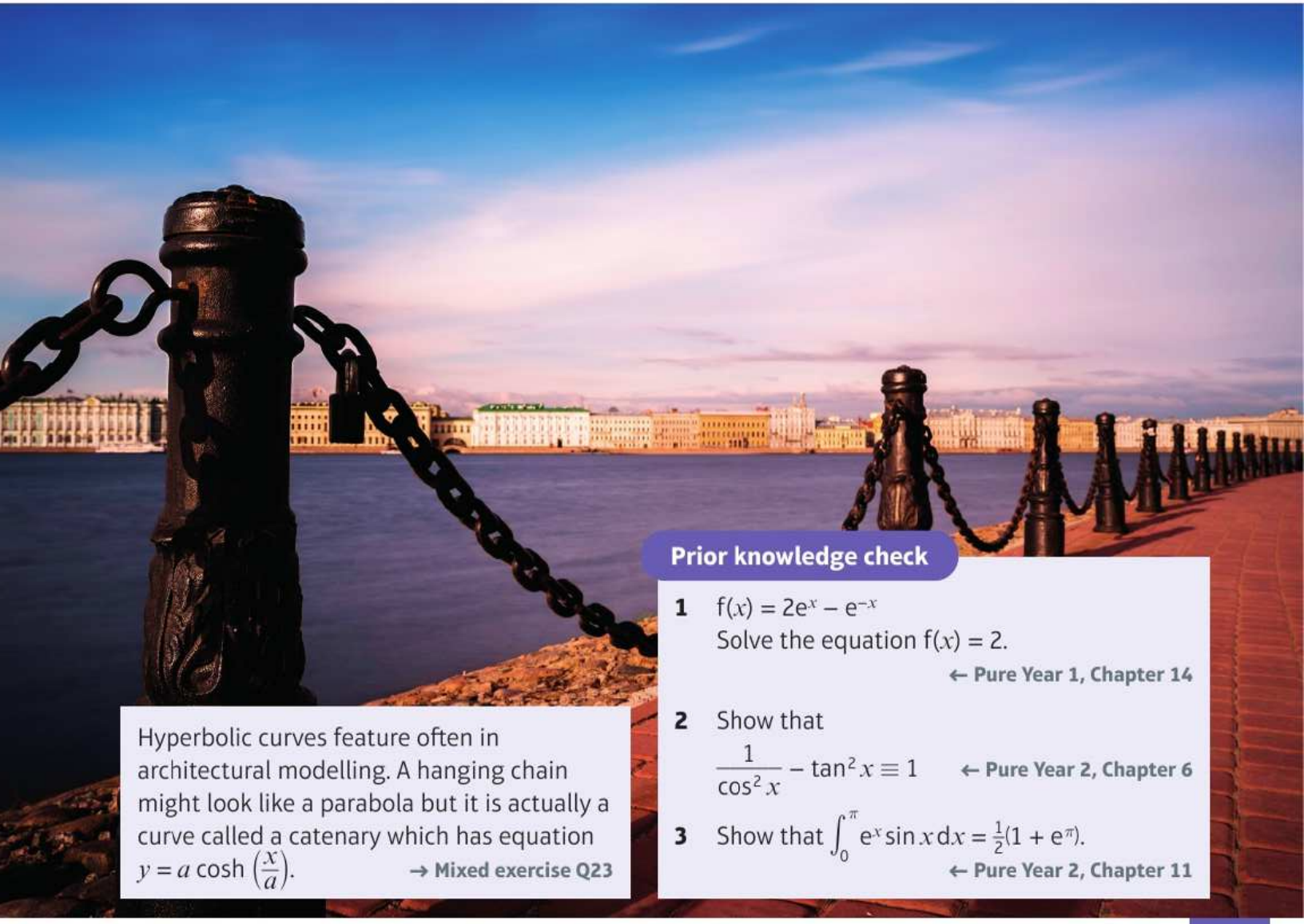
Hyperbolic functions

6

Objectives

After completing this chapter you should be able to:

- Understand the definitions of hyperbolic functions → pages 120–123
- Sketch the graphs of hyperbolic functions → pages 121–123
- Understand and use the inverse hyperbolic functions → pages 123–125
- Prove identities and solve equations using hyperbolic functions → pages 125–129
- Differentiate and integrate hyperbolic functions → pages 130–142



Hyperbolic curves feature often in architectural modelling. A hanging chain might look like a parabola but it is actually a curve called a catenary which has equation $y = a\cosh\left(\frac{x}{a}\right)$. → Mixed exercise Q23

Prior knowledge check

1 $f(x) = 2e^x - e^{-x}$
Solve the equation $f(x) = 2$.
← Pure Year 1, Chapter 14

2 Show that
$\frac{1}{\cos^2 x} - \tan^2 x \equiv 1$ ← Pure Year 2, Chapter 6

3 Show that $\int_0^{\pi} e^x \sin x \, dx = \frac{1}{2}(1 + e^{\pi})$.
← Pure Year 2, Chapter 11